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High Bandwidth Memory

What It Is and Why It Matters

High Bandwidth Memory (HBM) is a type of computer memory used in leading artificial intelligence (AI) chips and other hardware for high-performance computing. HBM can store and move large volumes of data quickly, making it suitable for increasingly data-intensive AI workloads.¹ As of 2026, most leading AI chips use HBM, and almost all leading AI models are trained and deployed on AI chips with HBM.² In the following section, we explain what sets HBM apart from other memory technologies and why HBM is important for AI's computational needs.

KEY TAKEAWAYS

- **Leading artificial intelligence (AI) workloads place extreme demands on memory, and High Bandwidth Memory (HBM) has emerged as the central technology to meet them.** Large AI models require chips with both high memory *capacity*—how much data can be stored—and high memory *bandwidth*—how fast data can be transferred to a processor. HBM is the only memory technology that meets both requirements: It offers higher bandwidth than other types of dynamic random-access memory (DRAM) while providing enough storage capacity to hold model weights and contextual data close to the processor, which is something that smaller, faster memory, such as static random-access memory, cannot do on its own. This combination is why most of the top AI chips use HBM.
- **HBM production is a complex, capital-intensive process that demands some of the world's most advanced manufacturing equipment.** First, companies fabricate individual DRAM memory *dies*—unpackaged silicon chips—at large production facilities. Each die requires etching nanometer-scale features using lithography tools that can cost more than \$300 million. The DRAM dies are then stacked vertically, which requires extreme precision because a single defect can compromise the entire HBM stack. HBM's complexity and resource-intensiveness make it expensive to produce: It can account for 45 percent of the total production cost of an AI chip, roughly three times the cost of fabricating the chip's processor.
- **Only three companies—SK Hynix (South Korea), Samsung (South Korea), and Micron (United States)—produce globally competitive HBM.** Their production is concentrated in South Korea, Taiwan, and Japan. Chinese company CXMT (ChangXin Memory Technologies) is attempting to increase its domestic HBM production but lags in overall output and does not manufacture the most-advanced HBM generations. In December 2024, China was barred from importing HBM from foreign producers, so Chinese access to advanced HBM depends on whether CXMT can improve.

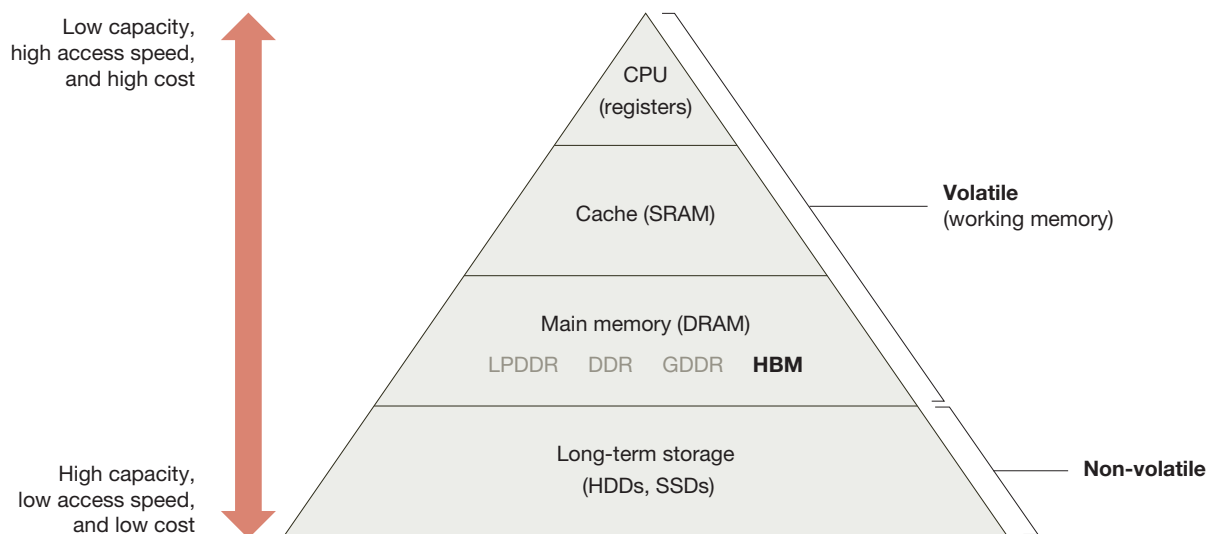
High Bandwidth Memory in the Broader Memory Landscape

The ability to store and access data is central to computing, but different workloads have varying requirements for how much memory is needed and how quickly they need to access that memory. Computer systems thus use a hierarchy of memory technologies that fall into two broad categories. The first is *volatile* memory: It requires constant power to retain data and serves as a system's active working memory, holding the data and instructions that running processes need.³ The second is *non-volatile* memory, which retains data without power and is used for long-term storage.⁴ In each category, there are multiple subtypes of memories that offer different trade-offs in access speed, capacity, and cost. In practice, a single device, whether a laptop or a server, relies on several of these technologies at once, each handling a different role. At one end of the spectrum are fast and expensive volatile memories with low capacity, such as static random-access memory (SRAM).⁵ SRAM is built into processors, such as central processing units (CPUs) or graphics processing units (GPUs), and provides data near-instantaneously, but it has

low capacity. At the other end are non-volatile storage types that are slow and cheap and offer high capacity. These include hard disk drives (HDDs) or solid-state drives (SSDs): They can hold large amounts of data affordably, but accessing that data takes a long time.

Dynamic random-access memory (DRAM) occupies a middle ground in capacity, bandwidth, and access speed (Figure 1). It is a volatile memory that has higher bandwidth and higher access speed than SSDs or HDDs and offers much greater capacity at a lower cost than SRAM.⁶ It comes in several variants optimized for different applications. Double data rate (DDR) DRAM is the standard, general-purpose variant, used in servers and personal computers; LPDDR (low-power DDR) is designed for energy efficiency in phones and laptops; and GDDR (graphics DDR) is optimized for graphics processing and offers higher bandwidth than standard DDR. HBM is the most specialized form of DRAM, designed to meet the extreme memory demands of leading AI workloads. Unlike conventional DRAM modules that place dies side by side on a circuit board, HBM stacks multiple DRAM dies vertically and places them next to the processor, providing greater memory capacity and higher data transfer rates in a smaller footprint.

FIGURE 1
High Bandwidth Memory in the Memory Hierarchy



NOTE: Computer systems organize memory into a hierarchy that trades off access speed, capacity, and cost. At the top of the figure sit the fastest and most expensive forms of memory per bit—CPU registers and cache (SRAM)—which hold only small amounts of data. Main memory (DRAM) occupies the middle tier, offering greater capacity at lower cost and speed. Within DRAM, several variants exist for different applications; HBM sits at the high-performance end, providing greater bandwidth than conventional DRAM types, such as LPDDR, DDR, and GDDR. Long-term storage memories (HDDs and SSDs) sit at the base. They have the highest capacity, lowest cost, and slowest access speeds. All working memory is volatile, meaning that data is lost when power is removed; only long-term storage retains data persistently.

Why Do Artificial Intelligence Workloads Need High Bandwidth Memory?

Leading AI workloads rely on specialized AI chips designed by such companies as Google, Nvidia, and Advanced Micro Devices (AMD).⁷ These chips deliver high processing speeds, but two things are required from memory to make effective use of that computational power: first, the ability to keep large amounts of data close to the processor, and second, high *bandwidth*—the rate at which data can move between memory and the processor.

Both AI training and inference require substantial memory capacity close to the processor. AI chips must hold a model's *parameters* (the numerical weights that determine the output of the model) in working memory so that the processor can access them quickly. Large AI models contain hundreds of billions of parameters, requiring hundreds of gigabytes (GBs) of memory just to hold the weights alone.⁸ For example, a 500-billion-parameter model in FP16 precision would need approximately 1 terabyte (TB) of memory, so much that the model needs to be distributed across at least six Nvidia B200 AI chips, each carrying 192 GB of HBM3E.⁹ Alternatively, the same model would need at least hundreds of SRAM-based chips to run because of their lower capacity per chip. Relying on long-term storage devices, such as HDDs or SSDs, would be impractical because they are orders of magnitude slower and cannot deliver data quickly enough to keep workloads running.

Model weights are not the only data that must reside in working memory. Modern AI models based on the transformer architecture also maintain what is known as a key-value (KV) cache. The KV cache functions like a notepad for everything the model has read so far in a given request so that the model does not have to reprocess the entire input each time it generates the next word. As the amount of input data a model can process (also known as the *context window*) has grown up to a million tokens, the KV cache has grown accordingly, consuming an increasing share of available memory alongside the model weights.¹⁰ Importantly, each concurrent user requires their own separate KV cache, leading to memory demands that scale along with the size of the user base.¹¹

Keeping all this data in memory is necessary but not sufficient: The processor also needs to read it fast enough to avoid sitting idle. During training and inference, massive volumes of data flow between memory and the AI chip's processor. If the data is not delivered fast enough, the processor stalls and the computing power goes unused.¹² In practice, much of the time spent generating new tokens during inference is not limited by how fast the processor can compute but by how fast it can read from memory. For those reasons, it is a *memory-bound* workload. This is why memory bandwidth is so critical for AI performance.

HBM provides a balance of specifications that meet AI's memory needs. It delivers far higher bandwidth than other types of DRAM, such as DDR or LPDDR, while offering enough dense capacity close to the processor to hold model weights and KV cache—something that on-chip SRAM, despite being faster, cannot provide on its own because of its limited capacity. No other single memory technology matches this combination. That said, as model sizes and context windows continue to grow, AI chip designers are actively adopting and exploring trade-offs across the memory hierarchy. These include allocating more die area to SRAM, pairing HBM with high-capacity DDR for less frequently accessed data, using SRAM-based chips alongside HBM-based designs for parts of inference, or developing new

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memory technologies. At the time of this writing in early 2026, HBM is the centerpiece, but the broader memory architecture around it continues to evolve.

High Bandwidth Memory Design

HBM's combination of high memory bandwidth and capacity makes it stand out from other memory alternatives, and both metrics have improved substantially across generations. The first generation of HBM debuted in 2013 and offered 128 GB per second of bandwidth and 4 GB of capacity per stack.¹³ HBM4, whose standard was finalized in 2025, delivers up to 2 TB per second of bandwidth and 64 GB of capacity per stack, increases of around 16 times for each.¹⁴

A typical AI chip contains multiple stacks of HBM (Figure 2).¹⁵ For example, Nvidia's B300, released in 2025, uses eight stacks of HBM3E. Each of the HBM stacks uses 12 DRAM dies and a base die, enabling the chip to reach 288 GB of total capacity and 8 TB per second of bandwidth.¹⁶

Two design features enable HBM to reach these performance specifications: vertical stacking and wide interfaces that can transmit high volumes of data in parallel. We explain both in detail in the following sections.

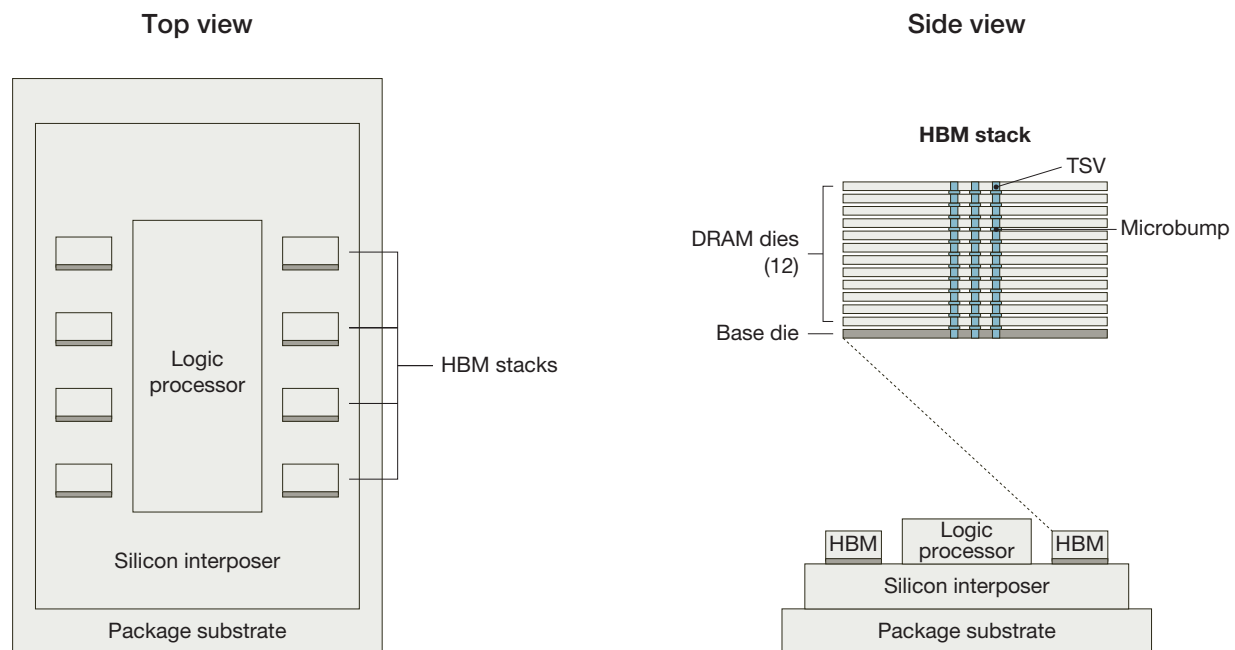
Vertical Stacking

A core feature of HBM is its vertical architecture: Multiple DRAM dies are vertically stacked (Figure 2) and thus substantially increase the amount of data HBM can store in a compact package. Microscopic vertical channels (TSVs) route data to and from individual dies in the stack at high throughput.¹⁷ At the bottom of the stack is a base logic die (base die), which orchestrates the rapid transfer of data between memory and the processor of the AI chip.

Wide Interfaces

Interfaces are physical communication pathways that allow components to exchange data. HBM has wide interfaces of 1024 bits (and 2048 bits in new

FIGURE 2
Artificial Intelligence Chip Using High Bandwidth Memory



SOURCE: Authors' adaptation from Ma and Patterson, "Challenges and Research Directions for Large Language Model Inference Hardware."

NOTE: This figure shows an AI chip with HBM. The left side of the figure (top view) displays several HBM stacks placed next to an AI chip's logic processor. Both are placed on a silicon interposer, which connects the HBM to the processor and is then attached to a package substrate. The right side of the figure (side view) displays the HBM's vertical stack, where 12 DRAM dies and a base die are connected with microscopic channels called through-silicon vias (TSVs).

HBM4 designs) that can move large quantities of data simultaneously into and out of memory storage.¹⁸ Because the channels are wide, they do not need extreme speeds to reach high bandwidths. This is a *wide and slow* design paradigm, which contrasts with a common alternative in which data is pushed at high rates through narrow channels (e.g., 64 bits in GDDR).¹⁹ One way to conceptualize HBM’s architecture is a wide, slow-moving river: Even if the water is not rushing quickly, the sheer width allows an enormous amount to flow through at once. The *wide and slow* approach offers another important advantage: Wide channels with lower speeds consume less power than narrower memory designs.²⁰

While HBM is well suited for AI, it has significant trade-offs, particularly its high production costs (an estimated \$2,900 for 192 GB of HBM3E, which is 45 percent of the cost to produce Nvidia’s B200 and three times the cost of producing its processor) and manufacturing complexity.²¹ We discuss this topic in the following section.

How Is High Bandwidth Memory Produced, and What Makes It So Challenging?

In the previous section, we explained HBM’s place in the broader memory landscape, why it is useful for leading AI workloads, and its main design features. In this section, we provide an overview of HBM’s manufacturing processes, which can be divided into three technically challenging domains, each consisting of many substeps: DRAM wafer fabrication, base die fabrication, and die stacking.

DRAM Wafer Fabrication

The HBM manufacturing process begins at DRAM fabrication facilities, known as *fabs* for short (Figure 3, Step 1). Here, manufacturers transform raw silicon wafers into functional memory dies through a series of steps that deposit, pattern, and remove materials on a nanometer scale. The fabrication process requires large clean rooms that span the size of multiple football fields and are outfitted with billions of dollars in specialized tools, including lithography machines—produced exclusively by the Dutch firm

ASML—that can cost more than \$300 million.²² As of 2026, front-end DRAM production for HBM is primarily in South Korea, Taiwan, and Japan, and there is more-limited capacity for lagging generations in China.²³ The United States does not have facilities that produce leading-edge DRAM for HBM at scale.

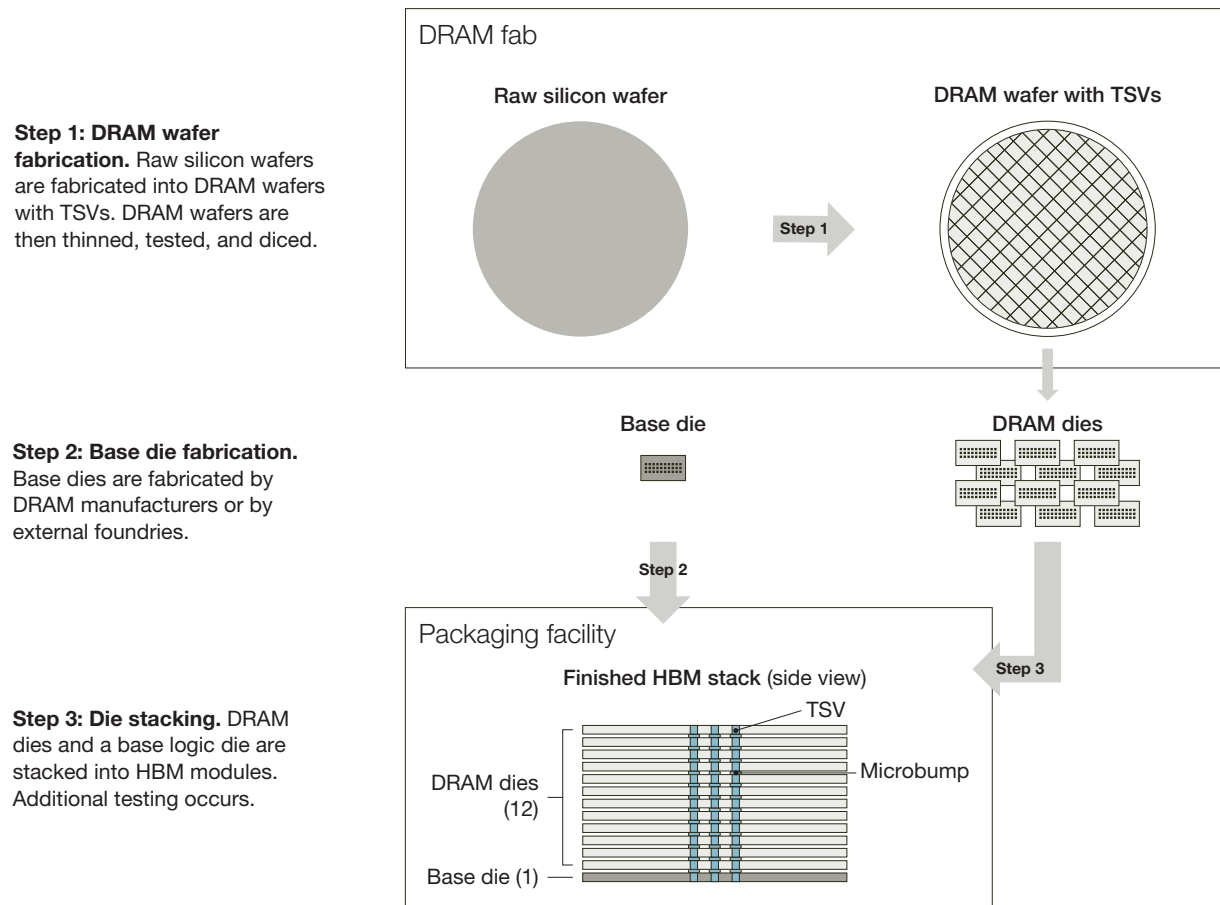
A key metric that determines a manufacturer’s DRAM sophistication is called the manufacturing *node*.²⁴ In semiconductor manufacturing, this metric measures the smallest feature size in nanometers (nm) that can be printed on a silicon wafer.²⁵ Smaller features are an important improvement because they increase memory cell density, allowing more bits (and thus more data) to be stored per unit of silicon area.²⁶ Importantly, these features also enable reductions in heat generation and energy consumption. Since producers started making DRAM with feature sizes below 20 nm in 2016, they have named nodes using the number “1” combined with a lettered progression (x, y, z, α , β , γ).²⁷ The big three memory manufacturers—Samsung, Micron, and SK Hynix—started volume production of 1z process nodes with feature sizes of around 16 nm between 2019 and 2020 (Figure 4).²⁸ As of 2026, they have advanced to feature sizes approaching 10 nm. Meanwhile, the most-advanced DRAM that Chinese memory manufacturer CXMT (ChangXin Memory Technologies) has produced at commercial scale in 2026 has a feature size of 16 nm, putting its technology around four years behind the industry’s state of the art.²⁹

Advances in DRAM are crucial for new HBM generations because improvements in DRAM density

While HBM is well suited for AI, it has significant trade-offs, particularly its high production costs and manufacturing complexity.

FIGURE 3

High Bandwidth Memory Manufacturing Process Flow



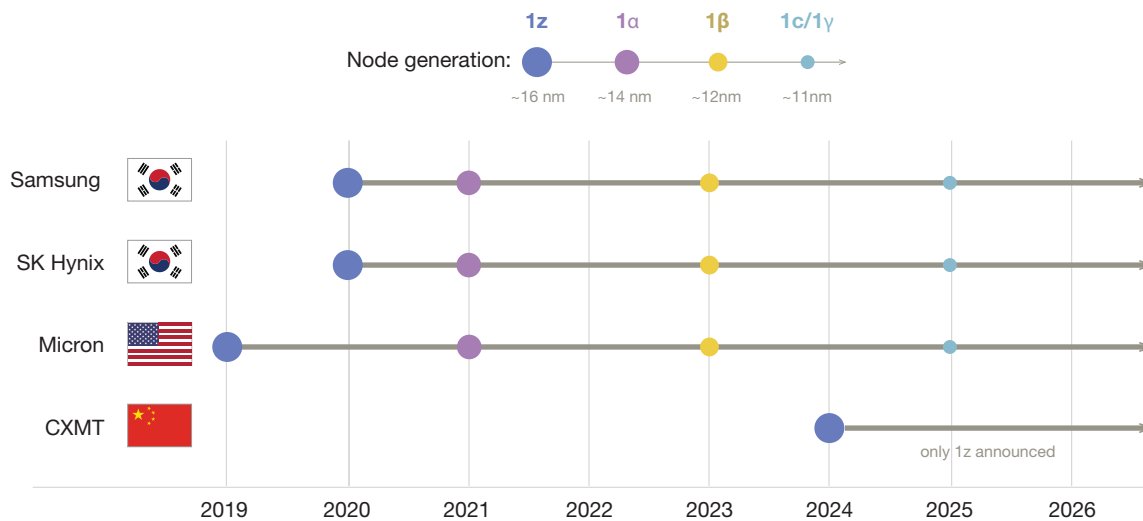
NOTE: Step 1: DRAM fabs transform raw silicon wafers into DRAM with TSVs in steps that deposit, pattern, and remove materials on a nanometer scale. DRAM wafers are then thinned, tested for defects, and diced into individual DRAM dies. Step 2: In parallel, companies also produce a base die that forms the foundation of the HBM stack. For generations up to HBM3E, the base dies are fabricated using DRAM processes. At HBM4 and beyond, some memory manufacturers are using logic processes. Step 3: Finally, DRAM dies and a base die are stacked together into an HBM stack.

help to unlock higher bandwidth and stack capacity.³⁰ Improvements also drive cost reductions for fabs, increasing the number of gigabytes that can be produced per wafer. However, migrating to more-advanced DRAM nodes is difficult and takes time: Manufacturers need to reoptimize certain process steps and introduce new materials and equipment.³¹ As a result, *manufacturing yields*—the percentage of dies that are functional on a silicon wafer—are lower when a new node is introduced but improve over time as problems are uncovered and resolved.³²

Unlike other types of memory dies, HBM requires DRAM with an extra step—the addition of TSVs used to route data through the dies of the vertical stack.³³ This increases the resource intensiveness

and difficulty of front-end manufacturing, for two main reasons. First, the DRAM dies used in HBM are larger and less dense than those in conventional DRAM because of the inclusion of TSVs.³⁴ As a result, in HBM, achieving the same memory capacity (in gigabytes) requires using significantly more silicon wafers—roughly three times as many as are needed for DDR5.³⁵ Second, forming the TSVs is a substantial technical challenge.³⁶ Each TSV requires drilling microscopic holes through the silicon, lining the holes with insulating material, and filling them with conducting metal; defects can occur at any of these steps.³⁷ TSV manufacturing is getting more challenging as HBM makers try to shrink the size of TSVs to preserve space for data storage.³⁸

FIGURE 4
Trends in DRAM Feature Size, by Manufacturer



SOURCES: Features data from Samsung Newsroom, “Samsung Begins Mass Production of 16Gb LPDDR5 DRAM at World’s Largest Semiconductor Line”; Samsung, “Samsung Starts Mass Production of Most Advanced 14nm EUV DDR5 DRAM”; Fan, “1c DRAM Key to Determining Memory Market Leadership in 2024”; “Samsung Reportedly Plans 200K 1c DRAM Wafers/Month by 2026, About One-Third of Its Total Output”; “SK Hynix Develops 1Znm 16Gb DDR4 DRAM”; Kostovic, “SK Hynix Starts Manufacturing 1anm DRAM Using EUV Lithography”; “SK Hynix Reportedly Set 1c DRAM Mass Production in Feb, While Samsung Delays by Six Months”; Bali, “Micron Commences Volume Production of 1z Nanometer DRAM Process Node”; Embedded Computing, “Micron Delivers Industry’s First 1-alpha DRAM Technology”; Lau, “Micron Ships World’s Most Advanced DRAM Technology with 1-Beta Node”; GFreeman, “Micron Ships World’s First 1γ (1-Gamma)-Based LPDDR5X, Enabling Rich Mobile AI Experiences”; “China Enters 2025 with Big Memory Breakthroughs.”

NOTE: * The size of the circles decreases with each new node generation (1z, 1α, 1β, 1c/1γ). ** Node generation sizes are approximations and vary by manufacturer. This figure shows advances in DRAM manufacturing for four memory manufacturers. Micron, Samsung, and SK Hynix reached volume production of roughly 16-nm feature sizes (the 1z node) between 2019 and 2020 and advanced to sub-12 nm-feature sizes in 2025. CXMT reached 16-nm feature sizes in 2024, putting the firm about four years behind the DRAM capabilities of Korean and U.S. manufacturers. Smaller feature sizes are better because they enable higher capacity and performance within a limited physical footprint.

HBM also requires that DRAM dies be thinned so that more layers can fit into an HBM stack. Thinning is a delicate step: Excessive stress can cause warping or TSV defects that reduce yield.³⁹ Finally, DRAM dies undergo testing to identify functional and defective units before they are diced, sorted, and prepared for stacking.

Base Die Fabrication

HBM requires a base die to manage data transfers between the stacked DRAM dies and the AI chip’s processor (Figure 3, Step 2). For HBM generations up to HBM3E, base dies for HBM were exclusively produced in house on DRAM process nodes. However, memory manufacturers are starting to migrate their base die fabrication to dedicated logic foundries, either at Samsung or Taiwan Semiconductor Manufacturing Company (TSMC).⁴⁰ Moving base die fabrication to logic foundries will contribute to

bandwidth and power efficiency gains, making the base die increasingly important for overall HBM performance.⁴¹ As of HBM4, base dies are increasingly customized to meet the design and performance objectives of a customer’s processor.⁴² They can be tuned to different priorities depending on the use case; for example, a power-constrained edge deployment might sacrifice bandwidth for a smaller, more efficient footprint, while a large-scale training cluster would optimize for maximum bandwidth and capacity regardless of power or size.⁴³

Base die fabrication is difficult because it combines the complexity and tool requirements of advanced logic with unique memory constraints. The die must accommodate high-speed interfaces and signaling to manage communication between the DRAM dies and the processor.⁴⁴ The base die also coordinates error correction functions across the stack, so defects can compromise the reliability of the entire module.⁴⁵ Because some base dies are

now manufactured using advanced logic processes, such as Samsung's 4-nm node and eventually TSMC's 3-nm "3NP" node, they require advanced lithography steps, including extreme ultraviolet (EUV) lithography at some layers.⁴⁶

Die Stacking

The final step in HBM manufacturing is bonding together the DRAM dies (Figure 3, Step 1) and base die (Figure 3, Step 2) into a vertical stack (Figure 3, Step 3). This process uses bonding technologies—typically thermocompression bonding or mass-reflow molded underfill (MR-MUF)—to connect dies using *micro-bumps*, tiny metal connections that are critical for the overall electrical and thermal performance of the HBM stack.⁴⁷ As stack heights progress to 16 dies and beyond, manufacturers expect to eventually adopt a technology called *hybrid bonding*, which stacks dies directly without the need for micro-bumps. This approach improves performance and power efficiency but introduces additional challenges, including high costs and difficulties related to yield, alignment, and warpage.

Notably, HBM's vertically stacked design creates a distinct yield problem: When the dies are stacked together, the defect rate compounds and an error in a single die can cause problems for the entire stack.⁴⁸ Here, the base die plays an important role because it helps route around defective parts of the DRAM dies. HBM's front-end and back-end complexity means producers commonly face production yields 20- to 30-percent lower than those for DDR5 memory.⁴⁹ For newer producers, such as CXMT, which are subject to manufacturing equipment restrictions from U.S. export controls and without the same years of accumulated learning, yields are expected to be even lower.⁵⁰

The Global High Bandwidth Memory Market

In the previous section, we explained the main steps involved in producing HBM and what makes this process so technically challenging. In this section, we provide an overview of the companies that produce

HBM as of 2026, explain where HBM is produced, and explore the factors shaping the HBM market.

A Small Number of Companies Are Competing for Market Share

Three companies produce advanced HBM at scale as of 2026: SK Hynix, Samsung, and Micron. Chinese memory giant CXMT has begun to produce lagging generations of HBM, but as of 2026, its output is small and it has yet to capture any measurable portion of global market share (Figure 5).⁵¹

SK Hynix leads globally by HBM market share, followed by Samsung and Micron.⁵² The company secured its lead because of its position as priority supplier of Nvidia GPUs, which accounted for more than 60 percent of total HBM consumption in both 2024 and 2025.⁵³ SK Hynix's development and use of MR-MUF bonding, which improves heat dissipation and overall yield, has been a key differentiator in achieving its dominant market position. The company was also the first to develop and mass-produce HBM in a 2014 partnership with AMD.⁵⁴ More than a decade of accumulated expertise has enabled SK Hynix to release new generations of HBM before its competitors, a compounding advantage that other companies have found difficult to overcome.⁵⁵

The world's largest memory manufacturer by DRAM capacity is Samsung, which is vying to win back HBM market share from SK Hynix.⁵⁶ Samsung was the primary supplier of HBM into China (a significant source of HBM demand) until December 2024, when the United States extended export controls to cover finished HBM units.⁵⁷ Samsung's chip unit has had a mixed record in winning customers since. In 2024 and 2025, it took the company 18 months after development was completed for Samsung's 12-layer HBM3E to meet Nvidia's qualification standards, a rigorous customer validation and testing process that suppliers must pass before their HBM products can be used in a customer's systems.⁵⁸ However, despite a period of reported challenges, Samsung supplied more than 60 percent of HBM for Google's tensor processing units—the second-most-produced AI chip in 2025. Samsung was also reportedly the first to ship commercial volumes of HBM4 in 2026.⁵⁹

FIGURE 5

Global High Bandwidth Memory Market Share, by Manufacturer



SOURCE: Features 2026 data from TrendForce on estimated market share.

NOTE: F = forecasted; Q = quarter. As of the fourth quarter of 2025, SK Hynix held 57 percent of the global HBM market by revenue. Samsung and Micron held 22 percent and 21 percent of the market, respectively. The Chinese memory manufacturer CXMT held a negligible global market share. Quarterly market share values for 2026 are forecasted, not observed.

The third company producing advanced HBM at scale as of 2026 is Micron. It is the only U.S.-headquartered HBM producer and held a market share of about 21 percent as of the end of 2025. Although it produces less HBM than Samsung or SK Hynix, Micron has secured major AI chip customers, including substantial volumes of HBM3E for Nvidia Blackwell platforms and AMD's MI350 chips.⁶⁰ The company began volume production of HBM4 designed for Nvidia's Vera Rubin in the first quarter of 2026.⁶¹

High Bandwidth Memory Production Is Centered in East Asia

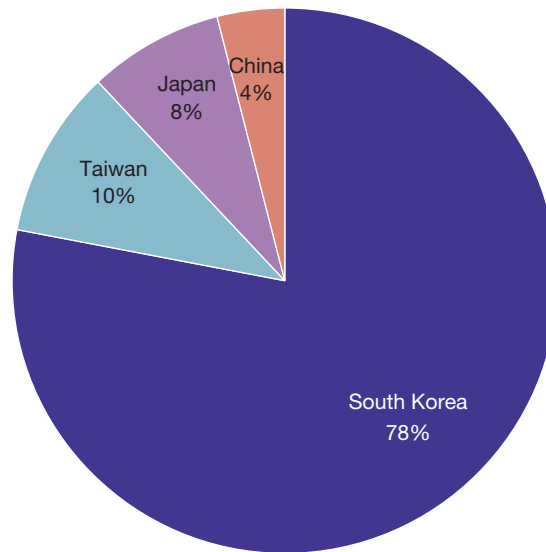
Geographically, HBM production is highly concentrated in East Asia. South Korea is estimated to be in control of 78 percent of HBM wafer capacity by the end of 2026, while Taiwan will have 10 percent, Japan will have 8 percent, and China will have 4 percent (Figure 6). As of this writing, Samsung Electronics and SK Hynix produce HBM exclusively in South Korea, while Micron manufactures in Taiwan

and Japan. Capacity expansions are reinforcing the region's central role: SK Hynix is developing a new semiconductor cluster in Yongin, South Korea, Samsung is expanding in Pyeongtaek, South Korea, and Micron is building or expanding facilities in Japan and in Singapore.

HBM's localized production reflects East Asia's deeply entrenched semiconductor ecosystem, which includes specialized suppliers, skilled labor, and proximity to advanced packaging capabilities, such as TSMC's Chip-on-Wafer-on-Substrate technology, which is used to package most leading AI chips. The United States does not produce HBM domestically and lacks the integrated ecosystem required for end-to-end production. Although there are no plans to fully localize HBM manufacturing in the United States, several relevant upstream and downstream investments are underway. SK Hynix is building a back-end HBM packaging facility in Indiana, which is expected to come online in 2028.⁶² Micron is expanding its front-end DRAM capacity in Boise, Idaho, while also developing a large-scale DRAM

FIGURE 6

Global High Bandwidth Memory Wafer Capacity Share, by Country



SOURCE: Features data from TrendForce on global HBM-TSV wafer capacity distribution.

NOTE: By the end of 2026, South Korea is projected to have 78 percent of global HBM wafer capacity, followed by Taiwan, Japan, and China.

fabrication complex in Clay, New York; production is expected to commence in 2030.⁶³ These investments could lay the groundwork for a more complete U.S. domestic capability over time, but barring significant changes in company planning, the United States will depend on South Korea, Japan, and Taiwan for HBM until at least 2030.

The Factors Shaping the High Bandwidth Memory Market

The HBM market is growing rapidly and is projected to expand from \$4 billion in 2023 to more than \$130 billion by 2033.⁶⁴ Beneath this growth trajectory are three structural factors that shape the HBM landscape regardless of which players lead: (1) high concentration with formidable barriers to entry, (2) customer demands driving rapid manufacturing process improvements, and (3) deliberate supply expansions.

High Bandwidth Memory Production Is Concentrated and Has High Barriers to Entry

Today, the HBM market is dominated by just three companies with the scale and technical prowess to compete. This concentration is the result of decades

of consolidation in the DRAM industry. In the 2000s, the market supported around ten major producers, but weaker players were forced to exit or merge when oversupply repeatedly crashed prices and cut profits.⁶⁵ By 2014, the top three memory manufacturers, Micron, Samsung, and SK Hynix, controlled around 90 percent of global DRAM market share, providing them with scale and process knowledge that later allowed them to dominate HBM.⁶⁶

Samsung, Micron, and SK Hynix's accumulated experience and production scale have created barriers that make entering the HBM market extremely challenging. The costs to bring new production online are staggering: A new, state-of-the-art fab can cost \$10 billion or more, which includes the construction of clean rooms and procurement of specialized equipment, such as deep ultraviolet (DUV) and EUV lithography, as well as bonding tools for etching TSVs.⁶⁷

Government incentives can help offset a portion of these costs, but these incentives often benefit incumbent companies rather than helping new companies enter the market. Much like advanced logic semiconductor manufacturing—where subsidies in Arizona or Japan primarily supported expansions by TSMC, Samsung, and Intel rather than new

entrants—state support for HBM production overwhelmingly flows to firms that already have scale, technical capabilities, and the ability to secure customer qualifications.⁶⁸

Even when substantial state backing is offered, technological barriers themselves remain formidable. For example, CXMT has received billions of dollars in financial support but does not yet produce leading HBM generations at large volumes.⁶⁹ Although capital can help a new entrant quickly expand production capacity, it cannot quickly compress the multiyear learning curves, research and development, and supplier integrations that shape competitiveness in HBM.

Demand Pressures Drive Manufacturing Improvements

With the introduction of each new HBM generation, manufacturers face customer pressure to improve performance (Figure 7).⁷⁰ The Joint Electron Device Engineering Council Solid State Technology Association (JEDEC), the industry’s standards body, sets baselines, but its specifications are typically conservative and consensus-based, designed so that multiple vendors can manufacture the products at scale. In practice, leading chip designers and hyperscalers push manufacturers to exceed JEDEC minimums, particularly for such metrics as per-pin transfer

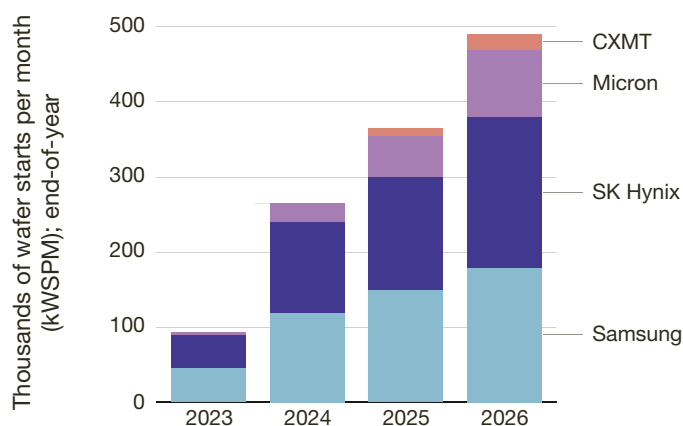
rates.⁷¹ Combined with increases in interface width and architectural improvements, these advances determine the total memory bandwidth achievable per HBM stack, and ultimately, per AI chip. For example, Nvidia’s Vera Rubin (VR200), in production as of 2026, uses HBM4 with around 20 TBs per second of memory bandwidth, six times the memory bandwidth of Nvidia’s H100, which began production in just 2022.⁷²

Commercial success in HBM depends on aligning production that meets aggressive customer roadmaps, but HBM’s technical difficulty makes this difficult in practice. Improving HBM performance requires either (1) advancing DRAM nodes or (2) adding more layers in an HBM stack, both of which come with significant challenges (see “How Is HBM Produced, and What Makes It So Challenging?”) and typically face lower yields at early production stages.⁷³ Competing in advanced HBM manufacturing thus involves rapidly switching to new processes and rapidly improving those processes’ reliability, which carries significant execution risk.

Supply Is Expanding Deliberately

In response to strong AI-driven demand, producers have expanded capacity steadily but deliberately, primarily by upgrading DRAM production lines with

FIGURE 7
High Bandwidth Memory Wafer Capacity, by Manufacturer



SOURCES: Features data from Wang, “How AI Growth Will Drive HBM Demand Beyond 2025”; Wu, “2024 HBM Bit Growth Estimated to Reach 260%, Making Up 14% of DRAM Industry”; and Wu, “2025 DRAM Market—Tight Supply Persists.”

NOTE: Global end-of-year HBM-TSV wafer capacity tripled between the end of 2023 and the end of 2025. SK Hynix leads in capacity, followed by Samsung, Micron, and CXMT. 2026 end-of-year values are forecasted.

Commercial success in HBM depends on aligning production that meets aggressive customer roadmaps, but HBM's technical difficulty makes this difficult in practice.

specialized equipment to produce HBM.⁷⁴ In absolute terms, total HBM wafer production capacity more than tripled between the end of 2023 and the end of 2025.⁷⁵ However, these expansions have been tightly tied to major customer roadmaps rather than more-speculative demand signals, creating the tight supply conditions observed in early 2026.⁷⁶

Two factors help explain why new supply has been relatively slow to come online despite fast-accelerating demand. First, meeting AI-driven demand for HBM requires more physical production capacity—more clean room space and more manufacturing tools. Manufacturers can add capacity by reallocating clean room space and tools used for other products, such as conventional DRAM and NAND flash storage, but beyond that, they must build entirely new facilities, which takes years and billions of dollars. Memory manufacturers are cautious about building this capacity, given the industry's long history of boom-and-bust cycles in which capacity built during strong demand periods comes online just as prices collapse.⁷⁷ Second, strict customer qualification requirements mean that added wafer capacity does not immediately translate into usable supply. Each new HBM generation must be validated for use in customer systems, sometimes creating monthslong delays between capacity additions and usable, commercial products.⁷⁸ The net

effect of these two factors is that tight supply periods are difficult to overcome on short time frames.

Conclusion

HBM is a central part of modern AI infrastructure. As AI models have scaled in size and complexity, HBM has been the only memory technology that is capable of delivering the combination of bandwidth and capacity that keeps pace with leading AI workloads. Other memory types might offer lower costs or greater capacity, but none match HBM's combined performance profile in a compact package close to the processor. Although AI systems and chip designs are actively evolving and incorporating alternative memory types, HBM is likely to retain a central role as AI models and inference demands continue to scale.

HBM's performance comes at a price, both technologically and economically. HBM can account for roughly 45 percent of the total cost of an AI chip, driven by a resource-intensive and complex manufacturing process that demands some of the most advanced semiconductor equipment, years of accumulated process expertise, and billions of dollars in capital. These barriers have kept the market narrow: As of 2026, only SK Hynix, Samsung, and Micron produce advanced HBM at globally competitive scale, and virtually all of it is manufactured in East Asia. CXMT, despite receiving significant financial and policy support, produces only low volumes of HBM that lag several generations behind.

For policy researchers, two areas are worth underscoring. First, given that U.S. export controls bar advanced HBM from export to China as of April 2026, how long it will take CXMT to narrow the gap with competitors will determine HBM access for China's AI chip makers. CXMT's ability to advance its HBM designs and production volumes will shape whether China's AI chipmakers remain bottlenecked or can secure a viable domestic source over time. Second, an open analytical question is whether U.S. industrial policy should aim to incentivize additional capacity to build a more integrated, end-to-end domestic HBM supply chain or whether the risk of relying on South Korean supply is sufficiently low

and East Asia's supplier ecosystem advantages sufficiently strong to make such measures unwarranted.

This paper has aimed to provide a foundation of what HBM is, how it is made, and who controls its production from which more targeted analysis can proceed. The authors welcome engagement from policymakers, researchers, and industry practitioners working on these questions.

Glossary

Artificial intelligence (AI) chip. An integrated circuit optimized for performing highly parallelized workloads, such as artificial intelligence training and inference. Also referred to as an artificial intelligence accelerator. Includes graphics processing units, tensor processing units, and other specialized processors.

ASML. A Dutch multinational corporation, it is the world's leading supplier of photolithography equipment, which is used in semiconductor manufacturing. ASML is the sole producer of extreme ultraviolet lithography machines and the dominant producer of deep ultraviolet lithography machines.

Cache. A small, fast memory layer located close to or directly on a processor that stores frequently accessed data to reduce the time and energy cost of retrieving them from slower memory. Caches are typically implemented using static random-access memory and are organized in hierarchical levels (e.g., L1, L2, L3). Lower levels are faster and smaller.

Central processing unit (CPU). A general-purpose processor that executes a wide variety of computing tasks and serves as the primary control unit in computer systems. CPUs are less specialized than artificial intelligence (AI) accelerators for AI workloads, but they are present in AI servers for handling non-AI tasks.

Changxin Memory Technologies (CXMT). A state-backed memory semiconductor manufacturer in China. CXMT is, as of 2026, the most advanced memory manufacturer in China.

Context window. The amount of information that an artificial intelligence model can recall when generating a response, measured in tokens.

Deep ultraviolet (DUV). A lithography technology that uses short wavelengths of light—typically 193 nanometers or 248 nanometers—to pattern semi-

conductor wafers. The most advanced form is DUV immersion lithography.

Die. An unpackaged semiconductor chip. Dies are fabricated on silicon wafers before they are cut and packaged.

Double data rate (DDR). The most widely used form of dynamic random-access memory, found in consumer devices and servers.

Dynamic random-access memory (DRAM). A type of volatile memory used to store data that is frequently accessed by the processor. DRAM must be continuously powered to retain data. High Bandwidth Memory is a specialized variant of DRAM.

Extreme ultraviolet (EUV). A lithography technology that uses 13.5-nanometer wavelengths of light to pattern semiconductor wafers at very fine resolutions. EUV enables advanced node fabrication and is used in leading-edge High Bandwidth Memory manufacturing. ASML is the only manufacturer of EUV machines.

Fabrication facility (fab). This is a term used in semiconductor manufacturing that refers to the large production facilities that produce semiconductors.

Graphics double data rate (GDDR). A variant of dynamic random-access memory that is optimized for graphics cards, offering higher bandwidth than standard double data rate. Used in graphics processing units for consumer and professional graphics workloads but with lower bandwidth than High Bandwidth Memory.

Graphics processing unit (GPU). A processor originally designed for rendering graphics, it has become the dominant chip for artificial intelligence training and inference workloads because of its ability to perform large numbers of mathematical operations in parallel.

Hard disk drive (HDD). A non-volatile storage device that offers large storage capacity at low cost but is slower than solid-state drives and far slower than dynamic random-access memory.

High Bandwidth Memory (HBM). A specialized variant of dynamic random-access memory (DRAM) that stacks multiple memory dies vertically and connects them with through-silicon vias. HBM delivers higher memory bandwidth and more capacity in a compact package than conventional DRAM and is

the standard memory technology used in advanced artificial intelligence chips.

Inference. The process of generating outputs from a trained artificial intelligence model. For example, when a user sends a message to a large language model and receives a response, that is inference.

Interposer. A thin silicon layer that provides high-density electrical connections needed to transfer data at high speed. The interposer is a key component of the advanced packaging that makes High Bandwidth Memory's performance possible.

Key-value cache (KV cache). A memory structure used during artificial intelligence inference that stores intermediate values from previously processed tokens, allowing the model to generate new tokens without recomputing prior context from scratch. The KV cache grows with context length and batch size and can reach terabytes in large-scale deployments.

Lithography. A semiconductor manufacturing process that uses light to transfer circuit patterns onto a silicon wafer. Lithography is one of the most critical steps in chip fabrication, determining the minimum feature size achievable on a given process node.

Logic process. A semiconductor fabrication process optimized for manufacturing digital logic circuits, including graphics processing units, central processing units, and networking processors.

Low-power double data rate (LPDDR). A variant of double data rate designed for energy efficiency, commonly used in mobile devices and laptops. LPDDR offers lower bandwidth than High Bandwidth Memory but consumes significantly less power.

Memory bandwidth. The rate at which data can be read from or written to memory, typically measured in gigabytes per second or terabytes per second. Memory bandwidth is a critical constraint in artificial intelligence systems.

Memory capacity. The total amount of data that memory can store at one time, typically measured in gigabytes. Distinct from memory bandwidth.

NAND. A high-density, non-volatile memory technology that retains data without power, commonly used in solid-state drives, USB drives, and smartphones.

Non-volatile memory. A type of memory that retains data without power.

Parameters. The numerical weights that define an artificial intelligence model's behavior, which are progressively refined during training.

Process node. A measure of miniaturization in semiconductor manufacturing, typically expressed in nanometers. Smaller nodes generally enable more transistors per chip and improved performance and energy efficiency, although cross-vendor comparisons can be misleading because node names do not always correspond directly with a physical dimension.

Processor. A semiconductor chip that executes computational instructions. Processors include central processing units, graphics processing units, and custom artificial intelligence (AI) accelerators. In AI systems, processors are typically paired with High Bandwidth Memory to meet the high memory bandwidth and capacity demands of training and inference workloads.

Registers. The smallest and fastest memory in a computer system, located directly within the processor. Registers hold the data and instructions that the processor is actively computing at any given moment.

Solid-state drive (SSD). A non-volatile storage device that uses flash memory to read and write data. SSDs are faster and more durable than hard disk drives but slower and less expensive than dynamic random-access memory.

Static random-access memory (SRAM). The fastest type of memory, typically located directly on a processor chip. SRAM delivers data near-instantaneously but is expensive and low-density, making it unsuitable for large-scale data storage.

Through-silicon via (TSV). A microscopic vertical electrical connection drilled through a silicon die. TSVs are used in High Bandwidth Memory stacks and enable extremely wide data pathways that give High Bandwidth Memory its high bandwidth.

Token. The smallest building blocks a model processes, such as a part of a word, image, or video.

Training. The process by which an artificial intelligence model learns by adjusting its parameters based on exposure to large amounts of data. Training is computationally intensive and places heavy simultaneous demands on both memory bandwidth and capacity.

Volatile memory. Memory that requires continuous power to retain data. Dynamic random-access memory, including High Bandwidth Memory, is volatile. When power is cut, all data stored in volatile memory is lost.

Wafer. A thin circular disc of silicon on which semiconductor chips are fabricated. Multiple dies are produced simultaneously on a single wafer before being cut apart.

Wafer starts per month (WSPM). The number of silicon wafers that begin the fabrication process at a given facility each month. WSPM is commonly used to estimate a fab's total production capacity.

Working memory. The volatile memory that holds data and instructions for use by the processor. Working memory keeps data close to the processor for fast access during computation, as opposed to long-term storage, which retains data persistently but is much slower to access.

Yield. The proportion of functional dies successfully produced from a wafer. Higher yields indicate more-efficient manufacturing with fewer defects. Low yields increase per-chip costs and constrain supply.

Notes

¹ Peddu, "AI's Rapid Growth"; Mujtaba, "Next-Gen HBM Architecture Detailed Including HBM4, HBM5, HBM6, HBM7 & HBM8"; Epoch AI, "Data on AI Models"; Heim and Lin, "When AI Takes Time to Think."

² Patel et al., "Scaling the Memory Wall"; Epoch AI, "Data on AI Models."

³ Computer Security Resource Center, "Glossary: Volatile Memory."

⁴ Awati and Sliwa, "Non-Volatile Memory (NVM)."

⁵ At the very top of the hierarchy sit *CPU registers*—tiny, ultrafast storage locations built directly into the processor core that hold values being actively computed at any given moment. Because registers are an internal part of the processor rather than a distinct memory technology, we do not cover them further in this paper.

⁶ Science Direct, "Dynamic Random Access Memory."

⁷ Sastry et al., "Computing Power and the Governance of Artificial Intelligence."

⁸ Epoch AI, "Data on AI Models."

⁹ Aubrey and Stam, "Inside NVIDIA Blackwell Ultra." FP16 is half-precision floating point, and HBM3E is HBM 3 Extended.

¹⁰ Jegou and Jeblick, "Mastering Long Contexts in LLMs with KVPress."

¹¹ Algorithmic optimizations, such as quantization techniques and more-efficient attention mechanisms, may reduce the memory capacity and bandwidth required per token. However, these approaches are best understood as part of a broader, continuous trajectory of efficiency improvements and are not necessarily discrete breakthroughs that fundamentally alter memory demand.

¹² Gholami et al., "AI and Memory Wall."

¹³ Ma and Patterson, "Challenges and Research Directions for Large Language Model Inference Hardware."

¹⁴ Ma and Patterson, "Challenges and Research Directions for Large Language Model Inference Hardware."

¹⁵ Patel et al., "Scaling the Memory Wall."

¹⁶ Aubrey and Stam, "Inside NVIDIA Blackwell Ultra."

¹⁷ Gambino, Adderly, and Knickerbocker, "An Overview of Through-Silicon-Via Technology and Manufacturing Challenges."

¹⁸ Ferro, "It's Official"; Kamath, "HBM4 Elevates AI Training Performance to New Heights"; Kingston, "DDR5 Memory Standard."

¹⁹ Ferro, "On the Road to Higher Memory Bandwidth."

²⁰ "The High-Stakes Race for Sustainable AI."

²¹ This production cost estimate from Epoch AI assumes 192 GB of HBM3E and a cost of between \$14 and \$17 per GB (Somala, "NVIDIA's B200 Costs Around \$6,400 to Produce, with Memory Accounting for Half").

²² A semiconductor clean room is a controlled environment that maintains low levels of airborne particles and stringent requirements for temperature or humidity (Chien, "Semiconductor Cleanrooms 101").

²³ Zakaria, "How Far Can Chinese HBM Go?"

²⁴ Notably, DRAM scaling has progressed at a slower pace than advanced logic, now needing roughly ten years to double bit density. Both the transistors and capacitors of a memory cell need to scale in tandem to enable bit density improvements, a paradigm that is approaching physical as chipmakers continue to push capacitor aspect ratios to the extreme (Patel et al., "The Memory Wall").

²⁵ More precisely, in DRAM, nodes track the "half-pitch" distance of the active area in a memory cell array (Tran, "Inside 1-Alpha DRAM, the World's Most Advanced DRAM Process Technology").

²⁶ "10,000 Times Thinner Than Human Hair?"

²⁷ "Samsung Starts Mass Producing Industry's First 10-Nanometer Class DRAM." Naming conventions also differ across memory manufacturers, and the same node does not imply equivalent technical specifications.

²⁸ Lee, Kim, and Lee, "SK Hynix Develops 1Znm 16Gb DDR4 DRAM"; "Samsung Develops Industry's First 3rd-Generation 10nm-Class DRAM for Premium Memory Applications"; Bali, "Micron Commences Volume Production of 1z Nanometer DRAM Process Node."

- 29 Choe, “First China-Made DDR5 Memory Released from CXMT China.”
- 30 For example, HBM3 uses 1z node DRAM, HBM3E uses the 1a node, and HBM4 is expected to rely on the 1β, 1c, or 1γ DRAM nodes (Morris and Wang, “Deep Dive on HBM”).
- 31 Varghese, “DRAM Scaling Requires New Materials Engineering Solutions.”
- 32 Terkel, “Yield Learning Curve Models in Semiconductor Manufacturing.”
- 33 Because TSV formation is the defining feature of front-end HBM production, HBM wafer capacity is often quoted in terms of *HBM-TSV* capacity, reflecting the fact that TSV processing ultimately defines output.
- 34 Patel et al., “Scaling the Memory Wall.”
- 35 James, “Here’s Why HBM Is Coming for Your PC’s RAM.”
- 36 Lee, Han, and Kang, “TSV Built-In Self-Repair Mechanism for Improving the Yield and Reliability of HBM.”
- 37 Peters, “What’s Next for Through-Silicon Vias.”
- 38 Kang, “Creating New Values in DRAM Using Through-Silicon Via Technology for Continued Scaling in Memory System Performance and Capacity.”
- 39 Applied Materials, “High Bandwidth Memory (HBM).”
- 40 Shilov, “HBM Undergoes Major Architectural Shakeup as TSMC and GUC Detail HBM4, HBM4E and C-HBM4E.”
- 41 Nomad76, “TSMC Showcases Custom C-HBM4E & N3P Logic Dies Target Double Efficiency.”
- 42 “Samsung Reportedly Moves Custom HBM Logic Die to 2nm Foundry Process for the First Time.”
- 43 Kanellos, “Custom HBM.”
- 44 Morris and Wang, “Deep Dive on HBM.”
- 45 Joo, “Samsung’s Development Phase for HBM4E Reaches Base Die Design, Custom HBM5 Next.”
- 46 “Samsung Ships Industry-First Commercial HBM4 with Ultimate Performance for AI Computing,” Samsung Newsroom; Shilov, “TSMC’s 3nm Update.”
- 47 Tsai and Han, “Bridging Performance and Yield.”
- 48 Sekar, “Why Is HBM So Hard to Manufacture?”
- 49 Clarke, “HBM to Be 20% of DRAM Market in 2024.”
- 50 According to commentary from TechInsights, “it will be difficult for [CXMT] to achieve a HBM3 mass production yield above 40% by the first half of next year” (Wang, “This is BIG if true . . .”).
- 51 Byung-Soo, “CXMT Struggles to Expand DRAM Production amid U.S. Sanctions.”
- 52 Other relevant ways to gauge competitiveness include evaluations of product, manufacturing, or customer integration metrics. At the product level, these could include per-pin data transfer rates or energy use per bit. On the manufacturing side, these could include cost per GB of capacity or cost per GB per second of bandwidth. Customer integration metrics, such as qualification breadth with leading AI chip designers, provide a practical signal of whether technical capabilities are translating into market adoption.
- 53 Wang, “How AI Growth Will Drive HBM Demand Beyond 2025.”
- 54 “Continuing to Make HBM History.”
- 55 Ahmad, “SK Hynix Maintains Memory Leadership with First HBM4.”
- 56 Wang, “How AI Growth Will Drive HBM Demand Beyond 2025.”
- 57 Patel et al., “Huawei AI CloudMatrix 384”; Liang, “China’s HBM Demands Account for 7% of 2024 Global Total.”
- 58 Hwang, Kim, and Park, “Samsung Clears Nvidia Hurdle for 12-Layer HBM3E Supply, Setting Stage for HBM4 Battle.”
- 59 “Samsung Ships Industry-First Commercial HBM4 with Ultimate Performance for AI Computing”; “Samsung Reportedly Supplies 60%+ of Google TPU HBM3E, Set to Remain Primary Supplier in 2026.”
- 60 “Micron Reveals Three Culprits Behind Memory Crunch”; “Micron Scales Up HBM to Four Major GPU/ASIC Clients, Targets 24% Market Share by Year-End.”
- 61 Liu Evensen and Kumar, “Micron in High-Volume Production of HBM4 Designed for NVIDIA Vera Rubin, PCIe Gen6 SSD and SOCAMM2.”
- 62 “The Anchor of Future AI Memory in Indiana.”
- 63 Newby and Kumar, “Micron Celebrates Official Ground-breaking at New York Megafab Site.”
- 64 Catsaros, “High-Bandwidth Memory Chip Market Could Grow to \$130 Billion by 2033, According to Bloomberg Intelligence.”
- 65 Patel et al., “Memory Mania.”
- 66 Bauer et al., “Memory.”
- 67 Mukano, “Micron to Invest \$9.6 Billion in Western Japan to Build AI Memory Chip Plant.”
- 68 For example, HBM manufacturing is designated a “national strategic technology” in South Korea; this designation confers substantial tax benefits and incentives to both Samsung and SK Hynix. The Japanese government has offered a variety of incentives to build memory fabs in Japan, including direct funding, tax breaks, infrastructure aid, and workforce development. In the United States, the 2022 Creating Helpful Incentives to Produce Semiconductors (CHIPS) Act extended direct funding, as well as tax and production credits, to Micron and SK Hynix (“Japan Reportedly Offers Half-Cost Fabs and Full Subsidies, yet Samsung and SK Hynix Stay Cautious”).
- 69 “With Two October Investments, Big Fund Phase II Commits Nearly 19 Billion RMB.”
- 70 Min-Gyu, “Nvidia Raises HBM4 Bar, Samsung and SK Hynix Race to Meet Next-Gen Chip Supply Standards.”
- 71 “HBM4 Race Heats Up.”

⁷² Goode, “Jensen Huang Says Nvidia’s New Vera Rubin Chips Are in ‘Full Production’”; AleksandarK, “NVIDIA Lowers HBM4 Specs for Vera Rubin VR200 as Memory Suppliers Miss 22 TB/s Target”; NVIDIA, “NVIDIA H100 Tensor Core GPU.”

⁷³ Terkel, “Yield Learning Curve Models in Semiconductor Manufacturing.”

⁷⁴ “Samsung and SK Hynix Urgently Reallocate 20% of DRAM Production Capacity to Support Rising HBM Demand.”

⁷⁵ Wang, “How AI Growth Will Drive HBM Demand Beyond 2025”; Wu, “2024 HBM Supply Bit Growth Estimated to Reach 260%, Making Up 14% of DRAM Industry.”

⁷⁶ Jae-Lim, “Samsung, SK Hynix Dramatically Scaling Up HBM Production to Meet Rising Demand, Recent Nvidia Deals.”

⁷⁷ The most recent example came during the coronavirus disease 2019 (COVID-19) pandemic, when manufacturers expanded aggressively to meet surging demand for consumer electronics, only to face severe oversupply and a price crash in 2022–2023.

⁷⁸ Lee, “Samsung Electronics Still Faces Challenges Securing HBM3E Orders Even If It Passes NVIDIA Quality Tests.”

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Abbreviations

AI	artificial intelligence
AMD	Advanced Micro Devices
CPU	central processing unit
CXMT	ChangXin Memory Technologies
DDR	double data rate
DRAM	dynamic random-access memory
DUV	deep ultraviolet (lithography)
EUV	extreme ultraviolet (lithography)
fab	fabrication facility
GB	gigabyte
GDDR	graphics double data rate
GPU	graphics processing unit
HBM	High Bandwidth Memory
HDD	hard disk drive
JEDEC	Joint Electron Device Engineering Council Solid State Technology Association
KV	key-value
LPDDR	low-power double data rate
MR-MUF	mass-reflow molded underfill
nm	nanometer
SRAM	static random-access memory
SSD	solid-state drive
TB	terabyte
TSMC	Taiwan Semiconductor Manufacturing Company
TSV	through-silicon via



About This Paper

Artificial intelligence (AI) systems depend on several enabling hardware technologies, and High Bandwidth Memory (HBM) is among the most critical. HBM is a specialized type of memory used in most leading AI chips, but it is produced at globally competitive scale by only three companies, and it relies on complex manufacturing processes concentrated in a handful of facilities in East Asia. In this paper, we synthesize technical and market analysis of the HBM ecosystem: how HBM works, how it is produced, and who controls its production. We aim to equip policymakers and researchers with a primer that can be referenced in a variety of policy discussions.

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